

# Extraction of the order- $e^0$ contribution to the scalar function $F_\phi$

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## Abstract

The Pauli–Villars regularized scalar current in Eq. (3.11) of Ref. [1], already contains an overall factor  $e^2$ . Consequently, its contribution of order  $e^2$  is obtained from the term of order  $e^0$  inside the square brackets. In this note I extract the corresponding constant term of the function  $F_\phi(\lambda, \mu)$  defined in Eq. (3.9), taking into account both  $\lambda = eE/H^2 = \mathcal{O}(e)$  and the implicit dependence  $\mu^2 = \mu_0^2 - \lambda^2$ . The apparent  $1/\lambda^2$  singularities cancel, and the remaining integral can be reduced analytically by means of the reflection identity for the digamma function.

## 1 Set-up

The regularized current is

$$\langle J_z^\phi \rangle_{\text{reg}} = aH \frac{e^2 E}{4\pi^2} \left[ \frac{1}{6} \log \left( \frac{\Lambda^2}{H^2} \right) - \frac{2\lambda^2}{15} + F_\phi(\lambda, \mu) \right], \quad (1.1)$$

where

$$\lambda = \frac{eE}{H^2}, \quad \mu^2 = \frac{9}{4} - (\bar{m}_\phi^2 + 12\xi) - \lambda^2, \quad \bar{m}_\phi = \frac{m_\phi}{H}. \quad (1.2)$$

It is convenient to define

$$\mu_0^2 \equiv \frac{9}{4} - (\bar{m}_\phi^2 + 12\xi), \quad \mu(\lambda) = \sqrt{\mu_0^2 - \lambda^2}.^1 \quad (1.3)$$

The perturbative expansion is performed for  $e \rightarrow 0$  while keeping  $E, H, \bar{m}_\phi, \xi$  and  $\Lambda$  fixed. Therefore

$$\lambda = \mathcal{O}(e), \quad \mu(\lambda) = \mu_0 - \frac{\lambda^2}{2\mu_0} + \mathcal{O}(\lambda^4), \quad (1.4)$$

where the intermediate expansion assumes  $\mu_0 \neq 0$ ; the final result can be continued to  $\mu_0 = 0$  by taking the appropriate limit. Defining the one-variable function obtained by following the physical curve, we get

$$\mathcal{F}(\lambda; \mu_0) \equiv F_\phi \left( \lambda, \sqrt{\mu_0^2 - \lambda^2} \right).$$

The first two terms of Eq. (3.9) are manifestly even in  $\lambda$ : the first contains  $\cosh(2\pi\lambda)/\lambda^2$ , whereas in the second both  $\sinh(2\pi\lambda)$  and  $\lambda^3$  change sign. For the integral, write its integrand as  $\mathcal{I}(r, \lambda)$ . It is not necessary that  $\mathcal{I}(r, -\lambda) = \mathcal{I}(r, \lambda)$  pointwise. Instead, all the nontrivial dependence occurs through  $r\lambda$  together with even powers of  $r$  and  $\lambda$ , so that

$$\mathcal{I}(-r, -\lambda) = \mathcal{I}(r, \lambda).$$

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<sup>1</sup>Solving  $\mu^2 = \mu_0^2 - \lambda^2$  algebraically gives  $\mu = \pm \sqrt{\mu_0^2 - \lambda^2}$ . Here the square root denotes the branch chosen continuously from  $\lambda = 0$ , so that  $\mu(0) = \mu_0$ .

Since  $r$  is a dummy integration variable and the interval  $[-1, 1]$  is symmetric, the change of variable  $r = -u$  gives

$$\begin{aligned} F_I(-\lambda) &= \operatorname{Re} \int_{-1}^1 dr \mathcal{I}(r, -\lambda) = \operatorname{Re} \int_{-1}^1 du \mathcal{I}(-u, -\lambda) \\ &= \operatorname{Re} \int_{-1}^1 du \mathcal{I}(u, \lambda) = F_I(\lambda). \end{aligned}$$

Thus  $\mathcal{F}(-\lambda; \mu_0) = \mathcal{F}(\lambda; \mu_0)$ , and its expansion contains only even powers:

$$F_\phi\left(\lambda, \sqrt{\mu_0^2 - \lambda^2}\right) = F_{\phi,0}(\mu_0) + \lambda^2 F_{\phi,2}(\mu_0) + \mathcal{O}(\lambda^4). \quad (1.5)$$

Here  $F_{\phi,0}$  is the constant coefficient and  $F_{\phi,2}$  is the coefficient of  $\lambda^2$ ; when  $\mathcal{F}$  is regular at the origin,  $F_{\phi,2} = \mathcal{F}''(0; \mu_0)/2$ . Moreover,  $\lambda = (E/H^2)e$  with  $E/H^2$  fixed. Therefore the constant coefficient in the expansion in  $e$  is the same as the constant coefficient in the expansion in  $\lambda$ :

$$[e^0]F_\phi = [\lambda^0]F_\phi = \lim_{\lambda \rightarrow 0} F_\phi\left(\lambda, \sqrt{\mu_0^2 - \lambda^2}\right). \quad (1.6)$$

The limit must be taken only after combining the terms of Eq. (3.9), because some of them are separately singular at  $\lambda = 0$ .

## 2 Expansion of $F_\phi$

The exact function in Eq. (3.9) is written below as the sum of three explicitly defined contributions:

$$\begin{aligned} F_\phi(\lambda, \mu) &= F_A(\lambda, \mu) + F_B(\lambda, \mu) + F_I(\lambda, \mu), \\ F_A(\lambda, \mu) &\equiv \frac{45 + 4\pi^2(-2 + 3\lambda^2 + 2\mu^2)}{12\pi^3} \frac{\mu \cosh(2\pi\lambda)}{\lambda^2 \sin(2\pi\mu)}, \\ F_B(\lambda, \mu) &\equiv -\frac{45 + 8\pi^2(-1 + 9\lambda^2 + \mu^2)}{24\pi^4} \frac{\mu \sinh(2\pi\lambda)}{\lambda^3 \sin(2\pi\mu)}, \\ F_I(\lambda, \mu) &\equiv \operatorname{Re} \left[ \int_{-1}^1 dr \frac{i}{16 \sin(2\pi\mu)} \right. \\ &\quad \times \left( -1 + 4\mu^2 + (7 + 12\lambda^2 - 12\mu^2)r^2 - 20\lambda^2 r^4 \right) \\ &\quad \times \left\{ (e^{-2\pi r\lambda} + e^{2\pi i\mu}) \psi\left(\frac{1}{2} + \mu - ir\lambda\right) \right. \\ &\quad \left. \left. - (e^{-2\pi r\lambda} + e^{-2\pi i\mu}) \psi\left(\frac{1}{2} - \mu - ir\lambda\right) \right\} \right]. \end{aligned} \quad (2.1)$$

Thus  $F_A$  and  $F_B$  denote the two elementary terms containing the hyperbolic functions, while  $F_I$  denotes the complete integral term.

### 2.1 Cancellation of the singular terms

Along the physical curve  $\mu^2 = \mu_0^2 - \lambda^2$ , introduce

$$\mathcal{C} \equiv 45 + 8\pi^2(\mu_0^2 - 1), \quad g(\mu) \equiv \frac{\mu}{\sin(2\pi\mu)}. \quad (2.2)$$

The two polynomial numerators simplify to

$$45 + 4\pi^2(-2 + 3\lambda^2 + 2\mu^2) = \mathcal{C} + 4\pi^2\lambda^2, \quad (2.3)$$

$$45 + 8\pi^2(-1 + 9\lambda^2 + \mu^2) = \mathcal{C} + 64\pi^2\lambda^2. \quad (2.4)$$

Substituting Eqs. (2.3)–(2.4) into the definitions of  $F_A$  and  $F_B$  in Eq. (2.1), the two elementary contributions, evaluated along the physical curve, can be written as

$$F_A(\lambda, \mu(\lambda)) = \frac{\mathcal{C} + 4\pi^2\lambda^2}{12\pi^3} g(\mu(\lambda)) \frac{\cosh(2\pi\lambda)}{\lambda^2},$$

$$F_B(\lambda, \mu(\lambda)) = -\frac{\mathcal{C} + 64\pi^2\lambda^2}{24\pi^4} g(\mu(\lambda)) \frac{\sinh(2\pi\lambda)}{\lambda^3}.$$

These identities are exact along  $\mu^2 = \mu_0^2 - \lambda^2$ ; no expansion in  $\lambda$  has yet been performed.

Moreover,

$$g(\mu(\lambda)) = g_0 + g_2\lambda^2 + \mathcal{O}(\lambda^4), \quad (2.5)$$

with

$$g_0 = \frac{\mu_0}{\sin(2\pi\mu_0)}, \quad g_2 = -\frac{1}{2\mu_0 \sin(2\pi\mu_0)} + \frac{\pi \cos(2\pi\mu_0)}{\sin^2(2\pi\mu_0)}. \quad (2.6)$$

The hyperbolic factors give

$$\frac{\cosh(2\pi\lambda)}{\lambda^2} = \frac{1}{\lambda^2} + 2\pi^2 + \mathcal{O}(\lambda^2), \quad (2.7)$$

$$\frac{\sinh(2\pi\lambda)}{\lambda^3} = \frac{2\pi}{\lambda^2} + \frac{4\pi^3}{3} + \mathcal{O}(\lambda^2). \quad (2.8)$$

We now display explicitly every product that can contribute through order  $\lambda^0$ . For  $F_A$ , substitution of Eqs. (2.5) and (2.7) gives

$$F_A = \frac{1}{12\pi^3} (\mathcal{C} + 4\pi^2\lambda^2) (g_0 + g_2\lambda^2 + \mathcal{O}(\lambda^4)) \left( \frac{1}{\lambda^2} + 2\pi^2 + \mathcal{O}(\lambda^2) \right).$$

The product of the first two parentheses is

$$(\mathcal{C} + 4\pi^2\lambda^2) (g_0 + g_2\lambda^2 + \mathcal{O}(\lambda^4)) = \mathcal{C}g_0 + (\mathcal{C}g_2 + 4\pi^2g_0)\lambda^2 + \mathcal{O}(\lambda^4).$$

Multiplying by the last parenthesis

$$F_A = \frac{1}{12\pi^3} \left[ \frac{\mathcal{C}g_0}{\lambda^2} + \mathcal{C}g_2 + 4\pi^2g_0 + 2\pi^2\mathcal{C}g_0 \right] + \mathcal{O}(\lambda^2).$$

Here, for example, terms such as  $2\pi^2(\mathcal{C}g_2 + 4\pi^2g_0)\lambda^2$  start at order  $\lambda^2$  and are therefore included in the remainder. After performing some simplifications one obtains

$$F_A = \frac{\mathcal{C}}{12\pi^3} \frac{g_0}{\lambda^2} + \frac{\mathcal{C}}{12\pi^3} g_2 + \left( \frac{\mathcal{C}}{6\pi} + \frac{1}{3\pi} \right) g_0 + \mathcal{O}(\lambda^2). \quad (2.9)$$

The same steps for  $F_B$  begin with

$$F_B = -\frac{1}{24\pi^4}(\mathcal{C} + 64\pi^2\lambda^2)(g_0 + g_2\lambda^2 + \mathcal{O}(\lambda^4)) \left( \frac{2\pi}{\lambda^2} + \frac{4\pi^3}{3} + \mathcal{O}(\lambda^2) \right).$$

The first two parentheses give

$$(\mathcal{C} + 64\pi^2\lambda^2)(g_0 + g_2\lambda^2 + \mathcal{O}(\lambda^4)) = \mathcal{C}g_0 + (\mathcal{C}g_2 + 64\pi^2g_0)\lambda^2 + \mathcal{O}(\lambda^4).$$

After multiplication by the hyperbolic expansion, the terms through order  $\lambda^0$  are

$$F_B = -\frac{1}{24\pi^4} \left[ \frac{2\pi\mathcal{C}g_0}{\lambda^2} + 2\pi\mathcal{C}g_2 + 128\pi^3g_0 + \frac{4\pi^3}{3}\mathcal{C}g_0 \right] + \mathcal{O}(\lambda^2).$$

Simplifying further

$$F_B = -\frac{\mathcal{C}}{12\pi^3} \frac{g_0}{\lambda^2} - \frac{\mathcal{C}}{12\pi^3} g_2 - \left( \frac{\mathcal{C}}{18\pi} + \frac{16}{3\pi} \right) g_0 + \mathcal{O}(\lambda^2). \quad (2.10)$$

Equations (2.9) and (2.10) show that both the  $1/\lambda^2$  poles and the  $g_2$  terms cancel. The constant contribution from the first two terms is therefore

$$\begin{aligned} [\lambda^0](F_A + F_B) &= \left( \frac{\mathcal{C}}{9\pi} - \frac{5}{\pi} \right) g_0 \\ &= \frac{8\pi\mu_0(\mu_0^2 - 1)}{9 \sin(2\pi\mu_0)}. \end{aligned} \quad (2.11)$$

## 2.2 Constant part of the integral

The integral term contains no negative power of  $\lambda$  after the two singular pieces have been kept separate as above. Its constant part is thus obtained by setting  $\lambda = 0$  and  $\mu = \mu_0$  in its integrand. We now make this substitution explicitly, separating all the factors that occur under the integral sign.

Let

$$\begin{aligned} P(r, \lambda, \mu) &\equiv -1 + 4\mu^2 + (7 + 12\lambda^2 - 12\mu^2)r^2 - 20\lambda^2r^4, \\ A_+(r, \lambda, \mu) &\equiv e^{-2\pi r\lambda} + e^{2\pi i\mu}, & A_-(r, \lambda, \mu) &\equiv e^{-2\pi r\lambda} + e^{-2\pi i\mu}, \\ \Psi_+(r, \lambda, \mu) &\equiv \psi\left(\frac{1}{2} + \mu - ir\lambda\right), & \Psi_-(r, \lambda, \mu) &\equiv \psi\left(\frac{1}{2} - \mu - ir\lambda\right). \end{aligned}$$

The quantity in braces in Eq. (2.1) is therefore

$$D(r, \lambda, \mu) \equiv A_+(r, \lambda, \mu)\Psi_+(r, \lambda, \mu) - A_-(r, \lambda, \mu)\Psi_-(r, \lambda, \mu),$$

and the complete integrand is

$$\mathcal{I}(r, \lambda, \mu) = \frac{i}{16 \sin(2\pi\mu)} P(r, \lambda, \mu) D(r, \lambda, \mu), \quad F_I(\lambda, \mu) = \text{Re} \int_{-1}^1 dr \mathcal{I}(r, \lambda, \mu).$$

Here  $\psi(z) = \Gamma'(z)/\Gamma(z)$  is the digamma function. Thus the object under the integral is the product of three pieces: the prefactor  $i/[16 \sin(2\pi\mu)]$ , the real polynomial  $P$ , and the generally complex combination  $D$ .

Along the physical curve  $\mu^2 = \mu_0^2 - \lambda^2$ , the polynomial can be written exactly as

$$\begin{aligned} P(r, \lambda, \mu(\lambda)) &= -1 + 4(\mu_0^2 - \lambda^2) \\ &\quad + \left[ 7 + 12\lambda^2 - 12(\mu_0^2 - \lambda^2) \right] r^2 - 20\lambda^2 r^4 \\ &= \underbrace{-1 + 4\mu_0^2 + (7 - 12\mu_0^2)r^2}_{P_0(r)} + \lambda^2(-4 + 24r^2 - 20r^4). \end{aligned}$$

Consequently its constant part is

$$P_0(r) = -1 + 4\mu_0^2 + (7 - 12\mu_0^2)r^2.$$

For the non-polynomial factors, setting  $\lambda = 0$  gives, one by one,

$$\begin{aligned} \mu(0) &= \mu_0, & e^{-2\pi r \lambda}|_{\lambda=0} &= 1, & -ir\lambda|_{\lambda=0} &= 0, \\ A_+(r, 0, \mu_0) &= 1 + e^{2\pi i \mu_0}, & A_-(r, 0, \mu_0) &= 1 + e^{-2\pi i \mu_0}, \\ \Psi_+(r, 0, \mu_0) &= \psi\left(\frac{1}{2} + \mu_0\right), & \Psi_-(r, 0, \mu_0) &= \psi\left(\frac{1}{2} - \mu_0\right). \end{aligned}$$

In particular, although  $D(r, \lambda, \mu)$  depends on  $r$  for nonzero  $\lambda$ , its constant part

$$\begin{aligned} D_0 &\equiv D(r, 0, \mu_0) \\ &= \left(1 + e^{2\pi i \mu_0}\right) \psi\left(\frac{1}{2} + \mu_0\right) - \left(1 + e^{-2\pi i \mu_0}\right) \psi\left(\frac{1}{2} - \mu_0\right) \end{aligned}$$

is independent of  $r$ . Therefore it may be pulled outside the  $r$  integral. The constant contribution is explicitly

$$\begin{aligned} [\lambda^0]F_I &= \text{Re} \left[ \int_{-1}^1 dr \frac{i}{16 \sin(2\pi \mu_0)} P_0(r) D_0 \right] \\ &= \text{Re} \left[ \frac{i D_0}{16 \sin(2\pi \mu_0)} \int_{-1}^1 dr P_0(r) \right]. \end{aligned}$$

This factorization is the reason that only the polynomial remains to be integrated. Its integral reduces to

$$\begin{aligned} \int_{-1}^1 dr \left[ -1 + 4\mu_0^2 + (7 - 12\mu_0^2)r^2 \right] \\ = 2(-1 + 4\mu_0^2) + \frac{2}{3}(7 - 12\mu_0^2) = \frac{8}{3}. \end{aligned} \quad (2.12)$$

Define

$$\psi_+ \equiv \psi\left(\frac{1}{2} + \mu_0\right), \quad \psi_- \equiv \psi\left(\frac{1}{2} - \mu_0\right), \quad s_0 \equiv \sin(2\pi \mu_0). \quad (2.13)$$

Equation (2.12) then gives

$$[\lambda^0]F_I = \frac{1}{6} \text{Re} \left[ \frac{i}{s_0} \left( (1 + e^{2\pi i \mu_0}) \psi_+ - (1 + e^{-2\pi i \mu_0}) \psi_- \right) \right]. \quad (2.14)$$

Writing  $c_0 = \cos(2\pi \mu_0)$ , the expression in parentheses becomes

$$(1 + c_0)(\psi_+ - \psi_-) + i s_0(\psi_+ + \psi_-). \quad (2.15)$$

To simplify the difference of the two digamma functions, begin with Euler's reflection formula for the Gamma function,

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

Taking the logarithm gives

$$\log \Gamma(z) + \log \Gamma(1 - z) = \log \pi - \log \sin(\pi z).$$

Recall that the digamma function is the logarithmic derivative of the Gamma function,

$$\psi(z) \equiv \frac{\Gamma'(z)}{\Gamma(z)} = \frac{d}{dz} \log \Gamma(z).$$

Differentiating the logarithmic reflection formula term by term, and using the chain rule in the second term, one finds

$$\begin{aligned} \frac{d}{dz} \log \Gamma(z) &= \psi(z), \\ \frac{d}{dz} \log \Gamma(1 - z) &= -\psi(1 - z), \\ \frac{d}{dz} [-\log \sin(\pi z)] &= -\pi \cot(\pi z). \end{aligned}$$

Consequently,

$$\psi(1 - z) - \psi(z) = \pi \cot(\pi z), \quad (2.16)$$

We now choose

$$z = \frac{1}{2} - \mu_0.$$

Then

$$1 - z = 1 - \left(\frac{1}{2} - \mu_0\right) = \frac{1}{2} + \mu_0,$$

so the left-hand side of Eq. (2.16) is precisely

$$\begin{aligned} \psi(1 - z) - \psi(z) &= \psi\left(\frac{1}{2} + \mu_0\right) - \psi\left(\frac{1}{2} - \mu_0\right) \\ &= \psi_+ - \psi_-. \end{aligned}$$

For the right-hand side,

$$\begin{aligned} \pi \cot(\pi z) &= \pi \cot\left[\pi\left(\frac{1}{2} - \mu_0\right)\right] \\ &= \pi \cot\left(\frac{\pi}{2} - \pi\mu_0\right) \\ &= \pi \tan(\pi\mu_0), \end{aligned}$$

where  $\cot(\pi/2 - x) = \tan x$  was used in the last step.

Therefore:

$$\psi_+ - \psi_- = \pi \tan(\pi\mu_0). \quad (2.17)$$

The trigonometric factor multiplying this difference can also be reduced explicitly. Since

$$c_0 = \cos(2\pi\mu_0), \quad s_0 = \sin(2\pi\mu_0),$$

give, for  $x = \pi\mu_0$ ,

$$\begin{aligned} \frac{1 + c_0}{s_0} &= \frac{1 + \cos(2\pi\mu_0)}{\sin(2\pi\mu_0)} \\ &= \frac{2 \cos^2(\pi\mu_0)}{2 \sin(\pi\mu_0) \cos(\pi\mu_0)} \\ &= \frac{\cos(\pi\mu_0)}{\sin(\pi\mu_0)}. \end{aligned}$$

Hence

$$\frac{1+c_0}{s_0} = \cot(\pi\mu_0), \quad (2.18)$$

It remains to substitute Eqs. (2.17) and (2.18) into Eq. (2.15). Denote the full parenthesis in Eq. (2.14) by

$$D_0 \equiv (1 + e^{2\pi i\mu_0})\psi_+ - (1 + e^{-2\pi i\mu_0})\psi_-.$$

Equation (2.15) states that

$$D_0 = (1 + c_0)(\psi_+ - \psi_-) + is_0(\psi_+ + \psi_-).$$

Multiplying both terms by the prefactor  $i/s_0$  gives

$$\begin{aligned} \frac{i}{s_0}D_0 &= \frac{i(1+c_0)}{s_0}(\psi_+ - \psi_-) + \frac{i}{s_0}is_0(\psi_+ + \psi_-) \\ &= i \cot(\pi\mu_0) \pi \tan(\pi\mu_0) - (\psi_+ + \psi_-) \\ &= i\pi - (\psi_+ + \psi_-). \end{aligned}$$

Restoring the definition of  $D_0$ , the result is

$$\frac{i}{s_0} \left( (1 + e^{2\pi i\mu_0})\psi_+ - (1 + e^{-2\pi i\mu_0})\psi_- \right) = i\pi - (\psi_+ + \psi_-). \quad (2.19)$$

Taking the real part removes the  $i\pi$  term, and hence

$$[\lambda^0]F_I = -\frac{1}{6} \operatorname{Re} \left[ \psi\left(\frac{1}{2} + \mu_0\right) + \psi\left(\frac{1}{2} - \mu_0\right) \right]. \quad (2.20)$$

### 3 Result

#### 3.1 The constant coefficient of $F_\phi$

Combining Eqs. (2.11) and (2.20), the term of order  $e^0$  of the function defined in Eq. (3.9) is

$$F_{\phi,0}(\mu_0) = \frac{8\pi\mu_0(\mu_0^2 - 1)}{9 \sin(2\pi\mu_0)} - \frac{1}{6} \operatorname{Re} \left[ \psi\left(\frac{1}{2} + \mu_0\right) + \psi\left(\frac{1}{2} - \mu_0\right) \right], \quad \mu_0^2 = \frac{9}{4} - (\bar{m}_\phi^2 + 12\xi). \quad (3.1)$$

This formula is best read as the sum of two conceptually distinct contributions,

$$F_{\phi,0}(\mu_0) = F_{\phi,0}^{\text{elem}}(\mu_0) + F_{\phi,0}^{\text{int}}(\mu_0), \quad (3.2)$$

where

$$F_{\phi,0}^{\text{elem}}(\mu_0) = \frac{8\pi\mu_0(\mu_0^2 - 1)}{9 \sin(2\pi\mu_0)}, \quad (3.3)$$

$$F_{\phi,0}^{\text{int}}(\mu_0) = -\frac{1}{6} \operatorname{Re} \left[ \psi\left(\frac{1}{2} + \mu_0\right) + \psi\left(\frac{1}{2} - \mu_0\right) \right]. \quad (3.4)$$

The first term comes from the two elementary, non-integral lines of Eq. (3.9), after their apparent  $1/\lambda^2$  singularities have cancelled. The second term is the constant part of the  $r$  integral.

The result is independent of the sign chosen for  $\mu_0$ . Indeed,  $\mu_0(\mu_0^2 - 1)$  and  $\sin(2\pi\mu_0)$  are both odd, so their ratio is even, while the two digamma functions are interchanged by  $\mu_0 \rightarrow -\mu_0$ . Consequently,

$$F_{\phi,0}(-\mu_0) = F_{\phi,0}(\mu_0), \quad (3.5)$$

as required because the mode equation depends on  $\mu$  only through  $\mu^2$ .

### 3.2 Why this produces the order- $e^2$ current

The key point is that the order of a term inside the square brackets of Eq. (1.1) is not yet the order of the full current. There is already an overall factor  $e^2 E$ . Along the physical curve,

$$\lambda^2 = \frac{e^2 E^2}{H^4} = \mathcal{O}(e^2), \quad F_\phi(\lambda, \mu(\lambda)) = F_{\phi,0}(\mu_0) + \lambda^2 F_{\phi,2}(\mu_0) + \mathcal{O}(\lambda^4). \quad (3.6)$$

It follows that the entire bracket has the expansion

$$\begin{aligned} & \frac{1}{6} \log\left(\frac{\Lambda^2}{H^2}\right) - \frac{2\lambda^2}{15} + F_\phi(\lambda, \mu(\lambda)) \\ &= \underbrace{\left[ \frac{1}{6} \log\left(\frac{\Lambda^2}{H^2}\right) + F_{\phi,0}(\mu_0) \right]}_{\text{order } e^0 \text{ inside the bracket}} + \lambda^2 \underbrace{\left[ F_{\phi,2}(\mu_0) - \frac{2}{15} \right]}_{\text{order } e^0} + \mathcal{O}(\lambda^4). \end{aligned} \quad (3.7)$$

Multiplication by the prefactor in Eq. (1.1) then gives

$$\begin{aligned} \langle J_z^\phi \rangle_{\text{reg}} &= aH \frac{e^2 E}{4\pi^2} \left[ \frac{1}{6} \log\left(\frac{\Lambda^2}{H^2}\right) + F_{\phi,0}(\mu_0) \right] \\ &\quad + aH \frac{e^4 E^3}{4\pi^2 H^4} \left[ F_{\phi,2}(\mu_0) - \frac{2}{15} \right] + \mathcal{O}(e^6). \end{aligned} \quad (3.8)$$

The first line is of order  $e^2$ ; the second line is of order  $e^4$ . Therefore the explicit term  $-2\lambda^2/15$  does not enter the order- $e^2$  current. Keeping only the leading term gives

$$\boxed{\langle J_z^\phi \rangle_{\text{reg}} = aH \frac{e^2 E}{4\pi^2} \left[ \frac{1}{6} \log\left(\frac{\Lambda^2}{H^2}\right) + F_{\phi,0}(\mu_0) \right] + \mathcal{O}(e^4).} \quad (3.9)$$

### 3.3 Dependence on mass and curvature coupling

All dependence of the extracted coefficient on the scalar mass and on the non-minimal curvature coupling is packaged into

$$X \equiv \overline{m}_\phi^2 + 12\xi, \quad \mu_0^2 = \frac{9}{4} - X. \quad (3.10)$$

The three useful regimes are

| Condition on $X$ | Value of $\mu_0$ | Mathematical form of the result |
|------------------|------------------|---------------------------------|
| $X < 9/4$        | real $\mu_0$     | trigonometric representation    |
| $X = 9/4$        | $\mu_0 = 0$      | limiting value of Eq. (3.1)     |
| $X > 9/4$        | $\mu_0 = i\rho$  | hyperbolic representation       |

For  $X > 9/4$ , writing  $\mu_0 = i\rho$  makes the reality of the coefficient explicit; this continuation is displayed in Eq. (4.35) below. The transition from real to imaginary  $\mu_0$  is therefore not a transition to a complex physical current. It is only a change in the most convenient parametrization of the solutions of the mode equation.



### 3.4 What has and has not been established

The calculation establishes the following facts:

- the apparent  $1/\lambda^2$  singularities of the elementary terms cancel;
- the exact function  $F_\phi$  has the finite constant coefficient  $F_{\phi,0}(\mu_0)$  along the physical curve  $\mu^2 = \mu_0^2 - \lambda^2$ ;
- this coefficient supplies the complete contribution of  $F_\phi$  to the regularized current at order  $e^2$ ;

## 4 Checks

### 4.1 Analytical checks

We consider the constant term

$$F_{\phi,0}(\mu_0) = \frac{8\pi\mu_0(\mu_0^2 - 1)}{9\sin(2\pi\mu_0)} - \frac{1}{6} \operatorname{Re} \left[ \psi\left(\frac{1}{2} + \mu_0\right) + \psi\left(\frac{1}{2} - \mu_0\right) \right], \quad (4.1)$$

where

$$\mu_0^2 = \frac{9}{4} - (\overline{m}_\phi^2 + 12\xi). \quad (4.2)$$

#### 4.1.1 Massless conformally coupled scalar

For a massless conformally coupled scalar,

$$\overline{m}_\phi = 0, \quad \xi = \frac{1}{6}, \quad (4.3)$$

and therefore

$$\mu_0^2 = \frac{9}{4} - 12\left(\frac{1}{6}\right) = \frac{1}{4}. \quad (4.4)$$

We choose  $\mu_0 = 1/2$ . Direct substitution into Eq. (4.1) is not allowed, since  $\sin(2\pi\mu_0) = 0$  and  $\psi(1/2 - \mu_0) = \psi(0)$  is singular. We therefore set

$$\mu_0 = \frac{1}{2} + \varepsilon, \quad \varepsilon \rightarrow 0. \quad (4.5)$$

We first consider the elementary contribution

$$F_{\phi,0}^{\text{elem}} = \frac{8\pi}{9} \frac{\mu_0(\mu_0^2 - 1)}{\sin(2\pi\mu_0)}. \quad (4.6)$$

The numerator is

$$\begin{aligned} \mu_0(\mu_0^2 - 1) &= \left(\frac{1}{2} + \varepsilon\right) \left[ \left(\frac{1}{2} + \varepsilon\right)^2 - 1 \right] \\ &= \left(\frac{1}{2} + \varepsilon\right) \left( -\frac{3}{4} + \varepsilon + \varepsilon^2 \right) \\ &= -\frac{3}{8} - \frac{1}{4}\varepsilon + \frac{3}{2}\varepsilon^2 + \varepsilon^3. \end{aligned} \quad (4.7)$$

The denominator is

$$\begin{aligned}
\sin(2\pi\mu_0) &= \sin(\pi + 2\pi\varepsilon) \\
&= -\sin(2\pi\varepsilon) \\
&= -2\pi\varepsilon + \frac{4}{3}\pi^3\varepsilon^3 + \mathcal{O}(\varepsilon^5).
\end{aligned} \tag{4.8}$$

Consequently,

$$\begin{aligned}
F_{\phi,0}^{\text{elem}} &= \frac{8\pi - \frac{3}{8} - \frac{1}{4}\varepsilon + \mathcal{O}(\varepsilon^2)}{9 - 2\pi\varepsilon + \mathcal{O}(\varepsilon^3)} \\
&= \frac{8\pi}{9} \left[ \frac{3}{16\pi\varepsilon} + \frac{1}{8\pi} + \mathcal{O}(\varepsilon) \right] \\
&= \frac{1}{6\varepsilon} + \frac{1}{9} + \mathcal{O}(\varepsilon).
\end{aligned} \tag{4.9}$$

The digamma contribution is

$$F_{\phi,0}^{\text{dig}} = -\frac{1}{6} \text{Re} [\psi(1 + \varepsilon) + \psi(-\varepsilon)]. \tag{4.10}$$

Around  $z = 1$ , the digamma function satisfies

$$\psi(1 + z) = -\gamma_E + \zeta(2)z + \mathcal{O}(z^2), \tag{4.11}$$

and therefore

$$\psi(1 + \varepsilon) = -\gamma_E + \frac{\pi^2}{6}\varepsilon + \mathcal{O}(\varepsilon^2). \tag{4.12}$$

Using the recurrence relation

$$\psi(z + 1) = \psi(z) + \frac{1}{z} \tag{4.13}$$

with  $z = -\varepsilon$ , we find

$$\psi(-\varepsilon) = \psi(1 - \varepsilon) + \frac{1}{\varepsilon}. \tag{4.14}$$

Since

$$\psi(1 - \varepsilon) = -\gamma_E - \frac{\pi^2}{6}\varepsilon + \mathcal{O}(\varepsilon^2), \tag{4.15}$$

it follows that

$$\psi(-\varepsilon) = \frac{1}{\varepsilon} - \gamma_E - \frac{\pi^2}{6}\varepsilon + \mathcal{O}(\varepsilon^2). \tag{4.16}$$

Combining Eqs. (4.12) and (4.16),

$$\psi(1 + \varepsilon) + \psi(-\varepsilon) = \frac{1}{\varepsilon} - 2\gamma_E + \mathcal{O}(\varepsilon^2). \tag{4.17}$$

Hence

$$F_{\phi,0}^{\text{dig}} = -\frac{1}{6\varepsilon} + \frac{\gamma_E}{3} + \mathcal{O}(\varepsilon^2). \tag{4.18}$$

Adding Eqs. (4.9) and (4.18), the poles cancel:

$$\begin{aligned}
F_{\phi,0} &= \left( \frac{1}{6\varepsilon} + \frac{1}{9} \right) + \left( -\frac{1}{6\varepsilon} + \frac{\gamma_E}{3} \right) + \mathcal{O}(\varepsilon) \\
&= \frac{\gamma_E}{3} + \frac{1}{9} + \mathcal{O}(\varepsilon).
\end{aligned} \tag{4.19}$$

Therefore,

$$\boxed{F_{\phi,0} \left( \frac{1}{2} \right) = \frac{\gamma_E}{3} + \frac{1}{9} = \frac{1}{3} \left( \gamma_E + \frac{1}{3} \right)}. \tag{4.20}$$

The corresponding contribution to the dimensionless regularized current is

$$\boxed{\mathcal{J}_{\phi,\text{reg}} \supset \frac{\lambda}{4\pi^2} F_{\phi,0} \left( \frac{1}{2} \right) = \frac{\lambda}{12\pi^2} \left( \gamma_E + \frac{1}{3} \right)}. \tag{4.21}$$

### 4.1.2 Imaginary $\mu_0$

Consider now the regime

$$\bar{m}_\phi^2 + 12\xi > \frac{9}{4}. \quad (4.22)$$

In this case  $\mu_0^2 < 0$ , and we define

$$\mu_0 = i\rho, \quad \rho = \sqrt{\bar{m}_\phi^2 + 12\xi - \frac{9}{4}} > 0. \quad (4.23)$$

The elementary contribution becomes

$$\begin{aligned} F_{\phi,0}^{\text{elem}}(i\rho) &= \frac{8\pi(i\rho) [(i\rho)^2 - 1]}{9 \sin(2\pi i\rho)} \\ &= \frac{8\pi(i\rho)(-\rho^2 - 1)}{9i \sinh(2\pi\rho)} \\ &= \frac{-8\pi i\rho(1 + \rho^2)}{9i \sinh(2\pi\rho)} \\ &= -\frac{8\pi\rho(1 + \rho^2)}{9 \sinh(2\pi\rho)}. \end{aligned} \quad (4.24)$$

Here we used

$$\sin(ix) = i \sinh x. \quad (4.25)$$

Equation (4.24) is manifestly real.

For the digamma contribution, we use

$$\psi(z^*) = \psi(z)^*. \quad (4.26)$$

Since

$$\left(\frac{1}{2} + i\rho\right)^* = \frac{1}{2} - i\rho, \quad (4.27)$$

we have

$$\psi\left(\frac{1}{2} - i\rho\right) = \psi\left(\frac{1}{2} + i\rho\right)^*. \quad (4.28)$$

Therefore,

$$\begin{aligned} &\psi\left(\frac{1}{2} + i\rho\right) + \psi\left(\frac{1}{2} - i\rho\right) \\ &= \psi\left(\frac{1}{2} + i\rho\right) + \psi\left(\frac{1}{2} + i\rho\right)^* \\ &= 2 \operatorname{Re} \psi\left(\frac{1}{2} + i\rho\right). \end{aligned} \quad (4.29)$$

The digamma contribution is consequently

$$F_{\phi,0}^{\text{dig}}(i\rho) = -\frac{1}{3} \operatorname{Re} \psi\left(\frac{1}{2} + i\rho\right), \quad (4.30)$$

which is also real.

Combining Eqs. (4.24) and (4.30), we obtain

$$\boxed{F_{\phi,0}(i\rho) = -\frac{8\pi\rho(1 + \rho^2)}{9 \sinh(2\pi\rho)} - \frac{1}{3} \operatorname{Re} \psi\left(\frac{1}{2} + i\rho\right)}. \quad (4.31)$$

Both terms are real, and hence

$$\boxed{F_{\phi,0}(i\rho) \in \mathbb{R}}. \quad (4.32)$$

Thus, although the mode parameter  $\mu_0$  is imaginary in this regime, the induced current remains real.

## 4.2 Massless conformally coupled scalar

For  $\overline{m}_\phi = 0$  and  $\xi = 1/6$ , one has  $\mu_0 = 1/2$ . Although the two terms in Eq. (3.1) are separately singular, their sum is finite. Expanding around  $\mu_0 = 1/2$  gives

$$\lim_{\mu_0 \rightarrow 1/2} F_{\phi,0}(\mu_0) = \frac{\gamma_E}{3} + \frac{1}{9} = \frac{1}{3} \left( \gamma_E + \frac{1}{3} \right). \quad (4.33)$$

Consequently, the finite regularized-current contribution is

$$\frac{\lambda}{4\pi^2} F_{\phi,0} \left( \frac{1}{2} \right) = \frac{\lambda}{12\pi^2} \left( \gamma_E + \frac{1}{3} \right), \quad (4.34)$$

in agreement with the regularized-current part of the weak-field result in Eq. (3.18) of Ref. [1].

## 4.3 Imaginary $\mu_0$

When  $\mu_0 = i\rho$  with  $\rho > 0$ , Eq. (3.1) becomes the manifestly real expression

$$F_{\phi,0}(i\rho) = -\frac{8\pi\rho(1+\rho^2)}{9\sinh(2\pi\rho)} - \frac{1}{3} \operatorname{Re} \psi \left( \frac{1}{2} + i\rho \right). \quad (4.35)$$

This also verifies that the continuation to the large effective-mass regime does not generate an imaginary current.

## References

- [1] M. Bastero-Gil, P. B. Ferraz, A. Torres Manso, L. Ubaldi and R. Vega-Morales, *Classical constant electric fields and the Schwinger effect in de Sitter*, arXiv:2508.14973v2 [hep-ph] (2026).